Tara Karimzadeh Sabet

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Summary

Computational materials scientist specializing in data-driven, machine-learning-accelerated modeling of electronic properties and thermal transport in complex oxides on high-performance computing (HPC) systems. Combines first-principles electronic structure methods, including density functional theory (DFT), with machine-learning interatomic potentials (MLIPs) to build reproducible, high-throughput simulation workflows that scale materials prediction from one-off calculations to production-ready pipelines. Published in *npj Computational Materials* and *Communications Materials*; PhD in Materials Science expected Dec 2026.

Education

Carnegie Mellon University — *Pittsburgh, PA*
Doctor of Philosophy in Materials Science — *Expected Dec 2026*
- **Doctoral Thesis:** First Principles Study of Vibrational Modes and Heat Transport in High-Entropy Oxides
- **Thesis Advisor:** Prof. Ismaila Dabo

Carnegie Mellon University — *Pittsburgh, PA*
Master of Science in Materials Science and Engineering — *Jan 2025 – Dec 2025*
- **Selected Coursework:** Introduction to Machine Learning, Methods of Computational Materials Science, Quantum Chemistry (I & II)

Iran University of Science and Technology — *Tehran, Iran*
Bachelor of Science in Materials Engineering | Concentration: Materials Design — *Aug 2016 – Aug 2020*
- **Thesis:** Solution Combustion Synthesis of Iron Oxide Nanoparticles for Supercapacitors
- **Selected Coursework:** Design & Selection of Engineering Materials, Thermodynamics of Materials, Economics & Industrial Management

Skills

Materials Science	Quantum ESPRESSO, VASP, DFPT, Monte Carlo, ASE, ICET, Pymatgen, Phonopy, Ovito, VESTA, CrystalMaker
Machine Learning	GNNs, Deep Learning, Surrogate Modeling, Gaussian Process Regression, Random Forests, Generative AI, LLMs
Programming & Scripting	Python (Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, Plotly), Bash, C++, C#, MATLAB, HTML, LaTeX
Dev Tools & HPC	Git/GitHub, Docker, Linux OS, Jupyter, VS Code, PyCharm, AWS SageMaker, Slurm/PBS Schedulers, atomate2, AiiDa, Parallelization (MPI, OpenMP), GPU Acceleration (CUDA)
General/Engineering	SolidWorks, ANSYS Mechanical, ANSYS Granta Selector, Origin, Microsoft Office Suite (Proficient)
Languages	Persian (Native Speaker), English (Fluent), French (Intermediate), German (Intermediate), Dutch (Intermediate)

Work Experience

Carnegie Mellon University — *Pittsburgh, PA*
Research Assistant — *Jan 2025 – Present*
- Automated a Monte Carlo simulated-annealing cation-swap workflow to generate a Warren–Cowley short-range-order ladder across high-entropy configurations, enabling controlled study of clustering effects on phonon spectra in collaboration with experimental EELS/TEM teams.
- Developed a machine-learned interatomic potential (MLIP) pipeline (CHGNet, atomate2) to predict phonon properties and structural stability across disordered, high-dimensional compositional spaces, accelerating property prediction by orders of magnitude over direct DFT, and establishing benchmarked accuracy against first-principles ground truth for structure/property screening at scale.
- Implemented a phonon band-unfolding code (spectral weight method) from scratch to project MLIP-derived force constants onto the primitive rocksalt Brillouin zone and resolve true phonon dispersions obscured by supercell folding.
- Evaluated ballistic and hydrodynamic formulations of the Boltzmann transport equation against atomistic calculations to improve nanoscale heat-transport predictions for quantum and semiconductor device design.
- Co-authored peer-reviewed research and perspective papers in top-tier journals (*npj Computational Materials*, *Communications Materials*).

Carnegie Mellon University — *Pittsburgh, PA*
Teaching Intern | Methods of Computational Materials Science, Foundations of Materials Sci. & Eng. — *Jan 2025 – Dec 2025*
- Designed and led hands-on recitation sessions reinforcing core concepts and problem-solving across multiple topics.
- Assisted instructors with course development and preparation, including lectures, homework problems, solutions, and recitation materials.
- Held open office hours to support students with coursework, assignments, and coding/software exercises.
- Graded exams, quizzes, and homework/laboratory assignments, ensuring timely and detailed feedback.

Pennsylvania State University — *University Park, PA*
Teaching Assistant | Thermodynamics of Materials, Computational Materials Science — *Sep 2022 – May 2024*
- Conducted recitations and office hours to support student learning and problem-solving.
- Evaluated homework, laboratory assignments, and exams, providing detailed feedback to reinforce key concepts.

Center for Nanoscale Science, Materials Research Science & Engineering Center (MRSEC) *University Park, PA*

Research Assistant (MRSEC–CMU subcontract following relocation to CMU in Jan 2025.) *Aug 2021 – Present*

- Developed a high-throughput materials-design framework for thermoelectrics combining ab initio phonon calculations on HPC systems and Debye-based thermal conductivity modeling to predict transport properties in high-entropy wolframites.
- Built a structure generation and screening pipeline (ICET, Wang–Landau Monte Carlo) across 2,286+ candidate crystal structures, mapping configurational stability and formation-energy landscapes to identify synthesizable, thermodynamically stable compositions, directly analogous to stability/synthesizability screening in generative crystal structure prediction (CSP) workflows.
- Led the computational arm of multiple NSF-funded cross-disciplinary projects with experimental teams to advance energy research.
- Directed weekly group meetings, managing presentation schedules and documentation to keep multi-institution collaborations on track.

Iran Khodro Automotive Company *Tehran, Iran*

Summer Research Intern *Jul 2019 – Sep 2019*

- Conducted mechanical and quality tests on auto bodies and coatings in R&D lab via cross-cut, stone chip, gas chromatography, and colorimetry.
- Crafted a comprehensive handbook on pretreatment/electrodeposition processes, detailing chemical reactions and operational procedures.
- Compared spray versus immersion phosphating on auto body panels through mechanical testing, demonstrating superiority of immersion.

Publications & Presentations

Peer-Reviewed Publications

[1] A. Beardo, W. Chen, B. McBennett, **T. Karimzadeh Sabet**, E. Nelson, T. Culman, H. Kapteyn, J. Knobloch, M. Murnane, I. Dabo. "Nanoscale Confinement of Phonon Flow and Heat Transport". *npj Computational Materials* **11** (1), 172 (2025).

[2] R. A. Robinson, **T. Karimzadeh Sabet**, F. Marques dos Santos Vieira, S. S. I. Almishal, S. V. G. Ayyagari, R. Katzbaer, G. Di Gianluca, S. Sarker, P. R. Trinidad-Perez, J. P. Barber, S. Gelin, S. H. Lee, J. Hodges, R. E. Schaak, V. H. Crespi, V. Gopalan, N. Alem, C. M. Rost, J-P. Maria, I. Dabo, Z. Mao. "High-Entropy Design of Transition Metal Oxide Semiconductors with Ultra-Low Thermal Conductivity". *Communications Materials* **7** (1), 198 (2026).

Manuscripts in Preparation

[1] **T. Karimzadeh Sabet**, S. Gelin, I. Dabo. "Phonon Band Unfolding Reveals Zone-Boundary Scattering from Mass and Force-Constant Disorder in High-Entropy Oxides". (2026).

[2] **T. Karimzadeh Sabet**, S. V. G. Ayyagari, S. Gelin, N. Alem, I. Dabo. "Effects of Tuning Short-Range Order on Vibrational Spectra in High-Entropy Rocksalt Oxides". (2026).

[3] S. V. G. Ayyagari, **T. Karimzadeh Sabet**, S. Gelin, I. Dabo, N. Alem. "Local Nuances in Vibrational Properties of High Entropy Oxides Revealed by STEM-EELS". (2026).

[4] M. McDonough, **T. Karimzadeh Sabet**, A. Graves, R. Trice, G. Nanda Gopakumar, Q. Zhang, M. Hilse, J. Redwing, I. Dabo, S. Law. "Growth of Aluminum Bismide (AlBi) via Molecular Beam Epitaxy". (2026).

Conference Presentations

[1] Ultra-low heat conductivity by configurational entropy for thermoelectric conversion. T. Karimzadeh Sabet. *American Physical Society Global Physics Summit*, Anaheim (CA), Mar. 2025. **Oral Presentation**.

[2] Electrocaloric performance of high-entropy oxides from first principles. T. Karimzadeh Sabet. *Materials Research Society Spring Meeting & Exhibit*, Seattle (WA), Apr. 2024. **Invited Oral Presentation**.

[3] Electrocaloric performance of high-entropy perovskites from first principles. T. Karimzadeh Sabet. *American Physical Society March Meeting*, Minneapolis (MN), Mar. 2024. **Oral Presentation**.

[4] Electric polarization in high-entropy oxides for electrocaloric refrigeration. T. Karimzadeh Sabet. *American Physical Society March Meeting*, Las Vegas (NV), Mar. 2023. **Poster Presentation**.

[5] First-principles prediction of the phase stability of high-entropy oxides. T. Karimzadeh Sabet. *American Physical Society March Meeting*, Las Vegas (NV), Mar. 2023. **Oral Presentation**.

[6] Temperature-dependent compositional stability and electric polarization of high-entropy oxides in electrocaloric cooling applications. T. Karimzadeh Sabet. *Conference Across MRSEC-PREM Schools*, Columbus (OH), Nov. 2022. **Poster Presentation**.

Additional Training

Deep Learning Specialization: Neural Networks and Deep Learning; Improving Deep Neural Networks *Sep 2024 – Feb 2025*

DeepLearning.AI (Credentials: Z2GI1J9WJTBK, BXIKJSH2IY7J)

- Implemented neural networks from scratch using forward/backpropagation and vectorization; applied deep learning techniques and CNNs/RNNs in Python and TensorFlow.
- Developed expertise in hyperparameter tuning, regularization (L2, dropout), batch normalization, optimization (Momentum, RMSprop, Adam).

Generative AI with Large Language Models *Aug 2024 – Sep 2024*

DeepLearning.AI & Amazon AWS (Credential: HVN9VG3I5ZK1)

- Studied Transformer architectures, LLM training/fine-tuning, scaling laws, and deployment strategies using Python and AWS SageMaker.

Achievements

2026 **MSE Teaching Intern Award**, Carnegie Mellon University *Pittsburgh, PA*

2025 **Thank a Teacher Recognition for Impactful Teaching (student-nominated)**, Carnegie Mellon University *Pittsburgh, PA*

2023 **Materials Science and Engineering Travel Award**, Pennsylvania State University *University Park, PA*

2020 **Merit-Based Exemption from the Graduate Entrance Examination**, Iran University of Science & Technology *Tehran, Iran*

2020 **Top-Ranked Student in Engineering Materials Design Cohort**, Iran University of Science and Technology *Tehran, Iran*

2019 **Academically Outstanding Student Award**, Iran University of Science & Technology *Tehran, Iran*

2015 **1st & 2nd Phase Qualifier, 15th Iran National Physics Olympiad**, Young Scholars Club *Tehran, Iran*

Leadership & Service

Multicultural Engineering Graduate Association *University Park, PA*

Vice President *Apr 2024 – Dec 2024*

- Spearheaded "MEGA Works," a series of structured academic co-working sessions to provide STEM graduate students with a productive, well-resourced environment—complete with amenities—focused on peer accountability and work progress.
- Co-organized events and professional development workshops, managing end-to-end logistics to maximize attendance and engagement.
- Co-launched the Graduate Student Community Space, to establish a multi-functional campus hub for studying, networking, and socialization.

Multicultural Engineering Graduate Association *University Park, PA*

Secretary *Apr 2023 – Apr 2024*

- Facilitated executive board meetings to strategize, design, and execute technical/professional development workshops and networking events.
- Centralized organizational workflows by maintaining detailed records of meeting logs, logistics, event metrics, and actionable milestones.

Center for Engineering Outreach and Inclusion, Penn State College of Engineering *Salt Lake City, UT*

Career Fair Exhibitor *Nov 2023*

- Represented Penn State Graduate School's College of Engineering as a graduate track sponsor and promoted recruitment opportunities at the Society of Hispanic Professional Engineers (SHPE) National Convention in Salt Lake City.
- Served as an ambassador at the Advanced Degree Showcase, actively engaging with prospective applicants to promote graduate programs.

Penn State Eberly College of Science *University Park, PA*

Workshop Facilitator *Jul 2022 – Feb 2024*

- Designed activities and educated K-12 students on sustainability and materials science through MRSEC outreach activities, such as the life cycle analysis of cell phones on the costs and carbon emissions—from extracting raw materials to production and shipping.
- Facilitated the following workshops: Science-U (Make it Matter) July 2022 & 2023, Exploration-U (Bellefonte Family STEAM Night) November 2023, ENVISION (STEM Career Day Supporting Young Women) February 2023 & 2024.

International Education and Resource Network *Tehran, Iran*

Volunteer of the Youth Project *Nov 2013 – Jul 2014*

- Spearheaded efforts in a high-impact team project that earned recognition and selection from a competitive pool of participating groups to be showcased at the 21st iEARN International Conference in Buenos Aires, Argentina.
- Managed community outreach and global communications for the "Volunteer of Youth" initiative; coordinated fundraising initiatives for pediatric cancer patients as well as volunteer operations at nursing homes, orphanages, and specialized care facilities.

References available upon request.